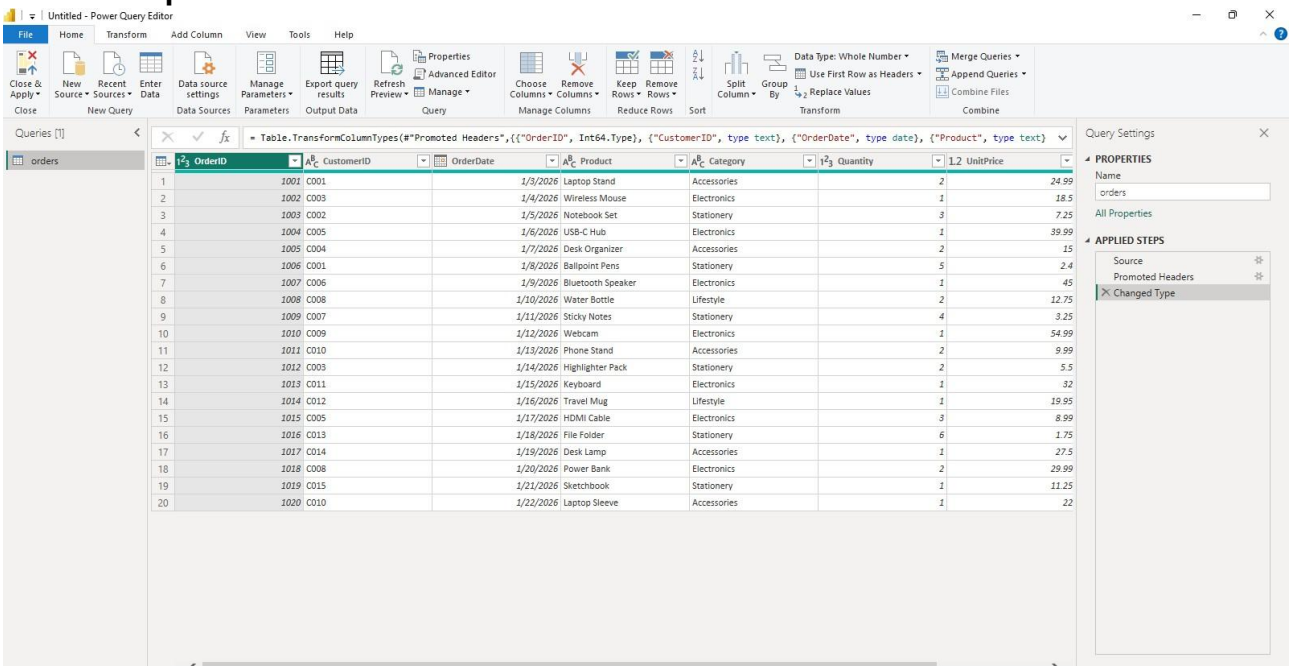
Assignment 1: Data Cleaning and Preparation using Power BI and KNIME

Datasets: orders.csv and customers.xlsx

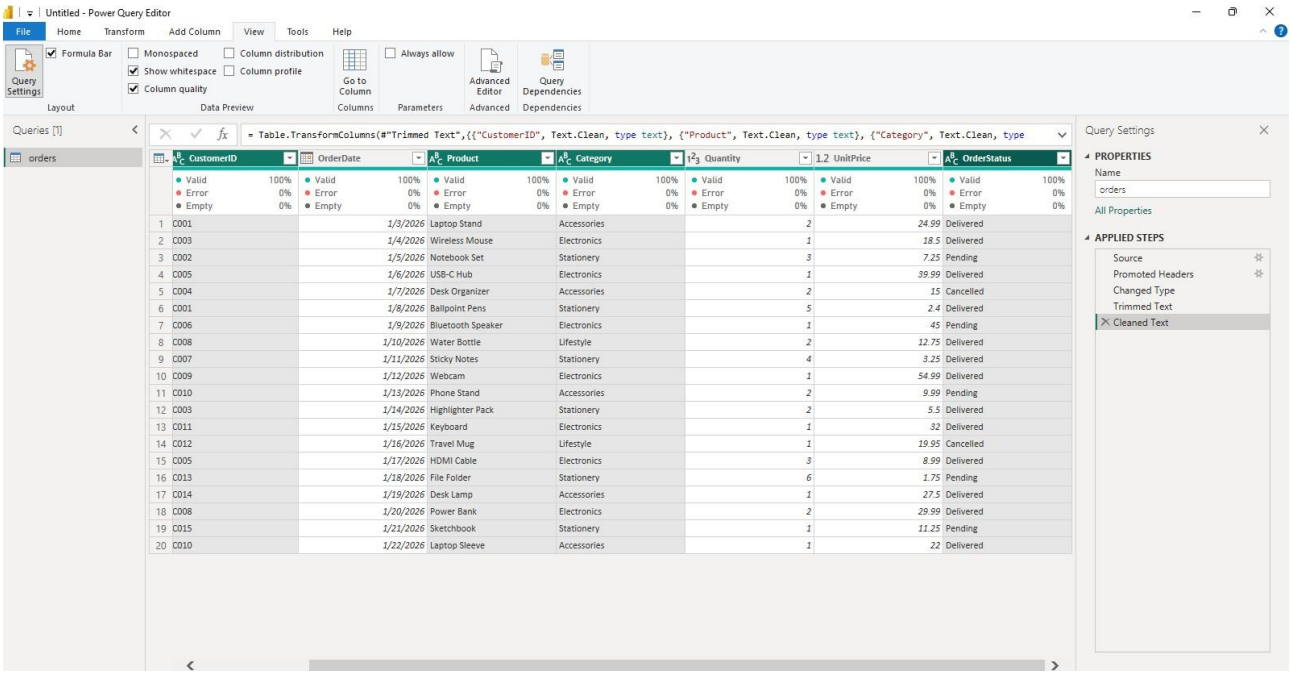
Task 1 : Import orders.csv



	OrderID	CustomerID	OrderDate	Product	Category	Quantity	UnitPrice
1	1001	C001	1/3/2026	Laptop Stand	Accessories	2	24.99
2	1002	C003	1/4/2026	Wireless Mouse	Electronics	1	18.5
3	1003	C002	1/5/2026	Notebook Set	Stationery	3	7.25
4	1004	C005	1/6/2026	USB-C Hub	Electronics	1	39.99
5	1005	C004	1/7/2026	Desk Organizer	Accessories	2	15
6	1006	C001	1/8/2026	Ballpoint Pens	Stationery	5	2.4
7	1007	C006	1/9/2026	Bluetooth Speaker	Electronics	1	45
8	1008	C008	1/10/2026	Water Bottle	Lifestyle	2	12.75
9	1009	C007	1/11/2026	Sticky Notes	Stationery	4	3.25
10	1010	C009	1/12/2026	Webcam	Electronics	1	54.99
11	1011	C010	1/13/2026	Phone Stand	Accessories	2	9.99
12	1012	C003	1/14/2026	Highlighter Pack	Stationery	2	5.5
13	1013	C011	1/15/2026	Keyboard	Electronics	1	32
14	1014	C012	1/16/2026	Travel Mug	Lifestyle	1	19.95
15	1015	C005	1/17/2026	HDMI Cable	Electronics	3	8.99
16	1016	C013	1/18/2026	File Folder	Stationery	6	1.75
17	1017	C014	1/19/2026	Desk Lamp	Accessories	1	27.5
18	1018	C008	1/20/2026	Power Bank	Electronics	2	29.99
19	1019	C015	1/21/2026	Sketchbook	Stationery	1	11.25
20	1020	C010	1/22/2026	Laptop Sleeve	Accessories	1	22

Summary: The orders.csv file is loaded into the Power Query Editor and shows 20 rows. The first row was promoted to headers and the data types were set, for example OrderID as whole number, OrderDate as date and UnitPrice as decimal. The Applied Steps pane shows Source, Promoted Headers and Changed Type.

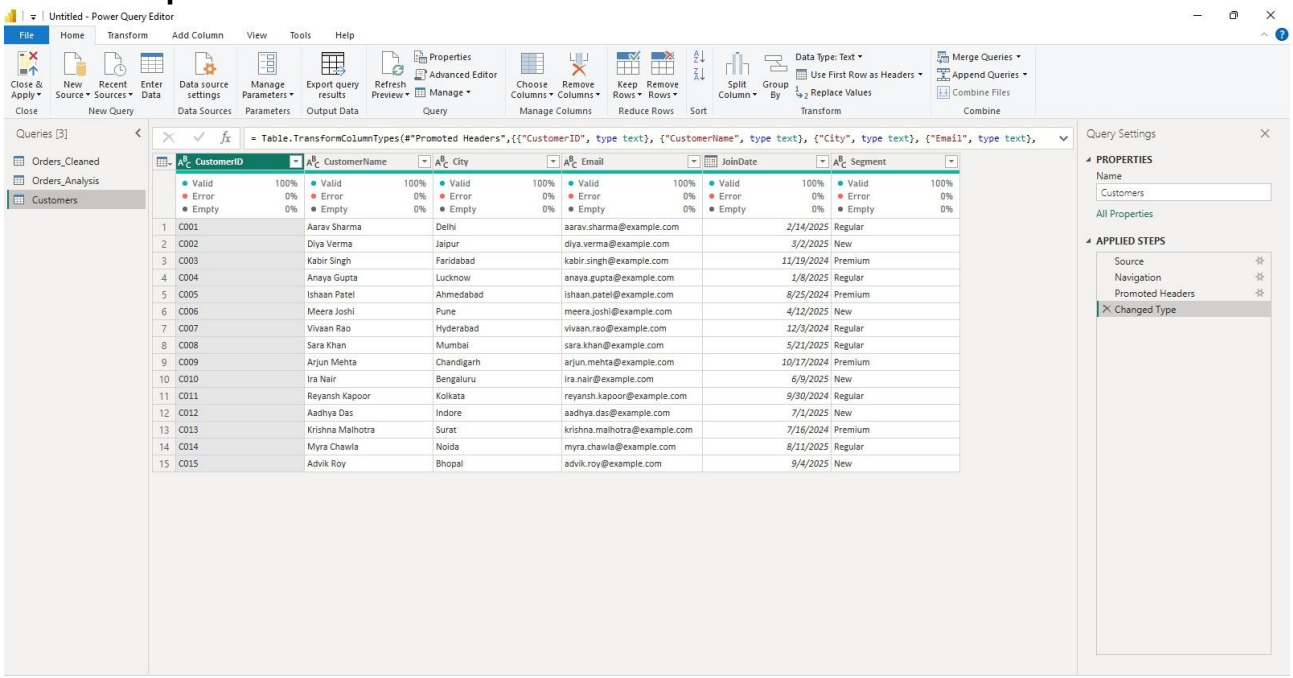
Task 2 : Clean the text columns



	CustomerID	OrderDate	Product	Category	Quantity	UnitPrice	OrderStatus
1	C001	1/3/2026	Laptop Stand	Accessories	2	24.99	Delivered
2	C003	1/4/2026	Wireless Mouse	Electronics	1	18.5	Delivered
3	C002	1/5/2026	Notebook Set	Stationery	3	7.25	Pending
4	C005	1/6/2026	USB-C Hub	Electronics	1	39.99	Delivered
5	C004	1/7/2026	Desk Organizer	Accessories	2	15	Cancelled
6	C001	1/8/2026	Ballpoint Pens	Stationery	5	2.4	Delivered
7	C006	1/9/2026	Bluetooth Speaker	Electronics	1	45	Pending
8	C008	1/10/2026	Water Bottle	Lifestyle	2	12.75	Delivered
9	C007	1/11/2026	Sticky Notes	Stationery	4	3.25	Delivered
10	C009	1/12/2026	Webcam	Electronics	1	54.99	Delivered
11	C010	1/13/2026	Phone Stand	Accessories	2	9.99	Pending
12	C003	1/14/2026	Highlighter Pack	Stationery	2	5.5	Delivered
13	C011	1/15/2026	Keyboard	Electronics	1	32	Delivered
14	C012	1/16/2026	Travel Mug	Lifestyle	1	19.95	Cancelled
15	C005	1/17/2026	HDMI Cable	Electronics	3	8.99	Delivered
16	C013	1/18/2026	File Folder	Stationery	6	1.75	Pending
17	C014	1/19/2026	Desk Lamp	Accessories	1	27.5	Delivered
18	C008	1/20/2026	Power Bank	Electronics	2	29.99	Delivered
19	C015	1/21/2026	Sketchbook	Stationery	1	11.25	Pending
20	C010	1/22/2026	Laptop Sleeve	Accessories	1	22	Delivered

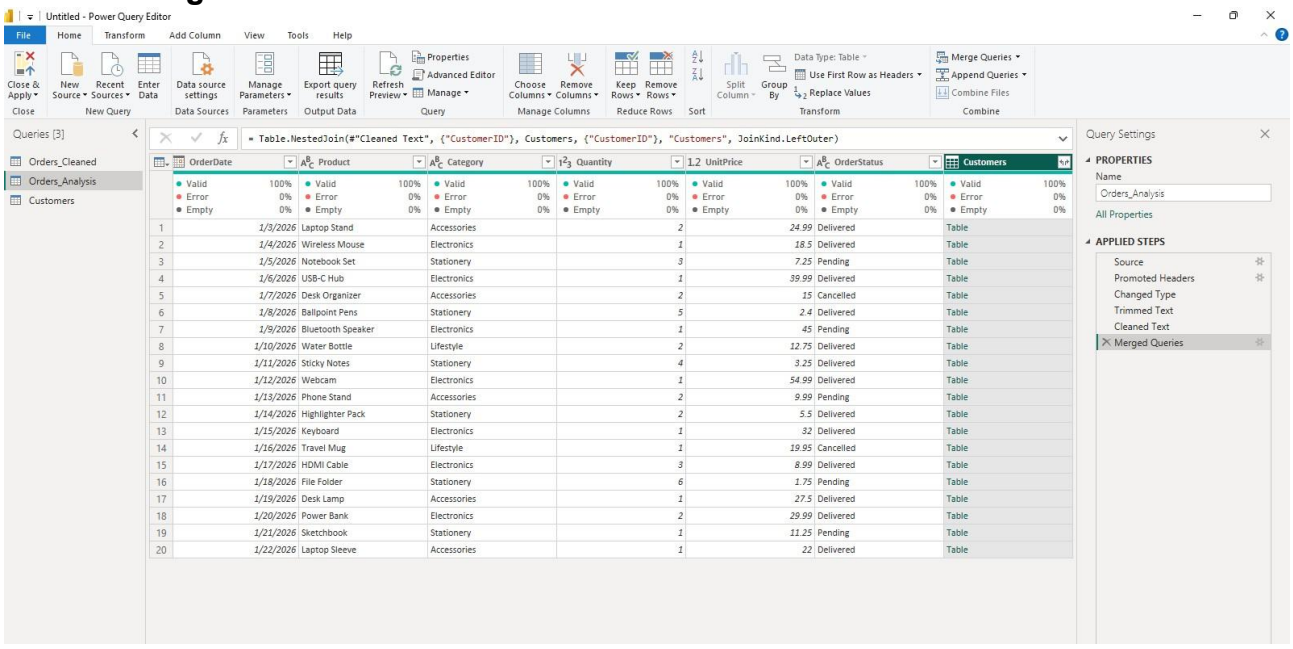
Summary: Trim and Clean steps were applied to the text columns to remove extra spaces and hidden characters. Column quality is turned on, and every column shows 100% valid, 0% error and 0% empty. This means the data is clean and has no missing values.

Task 3 : Import customers.xlsx



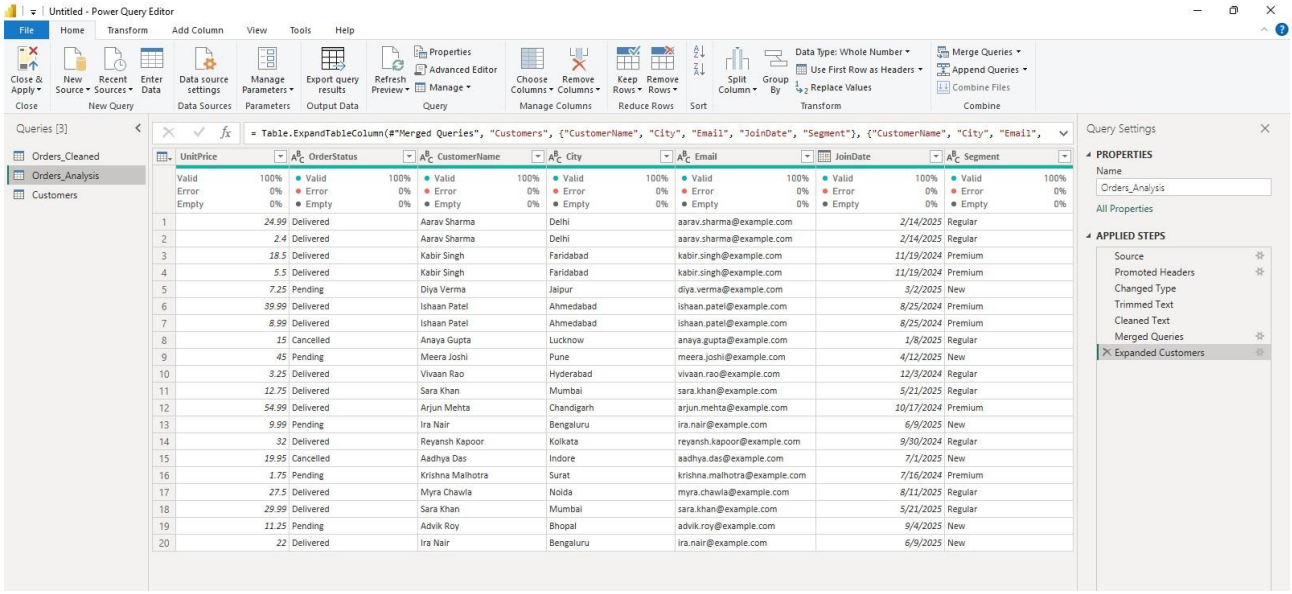
Summary: The customers.xlsx file is loaded as a new query named Customers. It has 15 customers with CustomerID, CustomerName, City, Email, JoinDate and Segment. Headers were promoted and data types were set, and all columns are 100% valid.

Task 4 : Merge orders and customers



Summary: The cleaned orders query was merged with the Customers query using CustomerID as the common column. A Left Outer join was used so all 20 orders are kept. A new Customers column appears, and each cell holds a table with that customer's details.

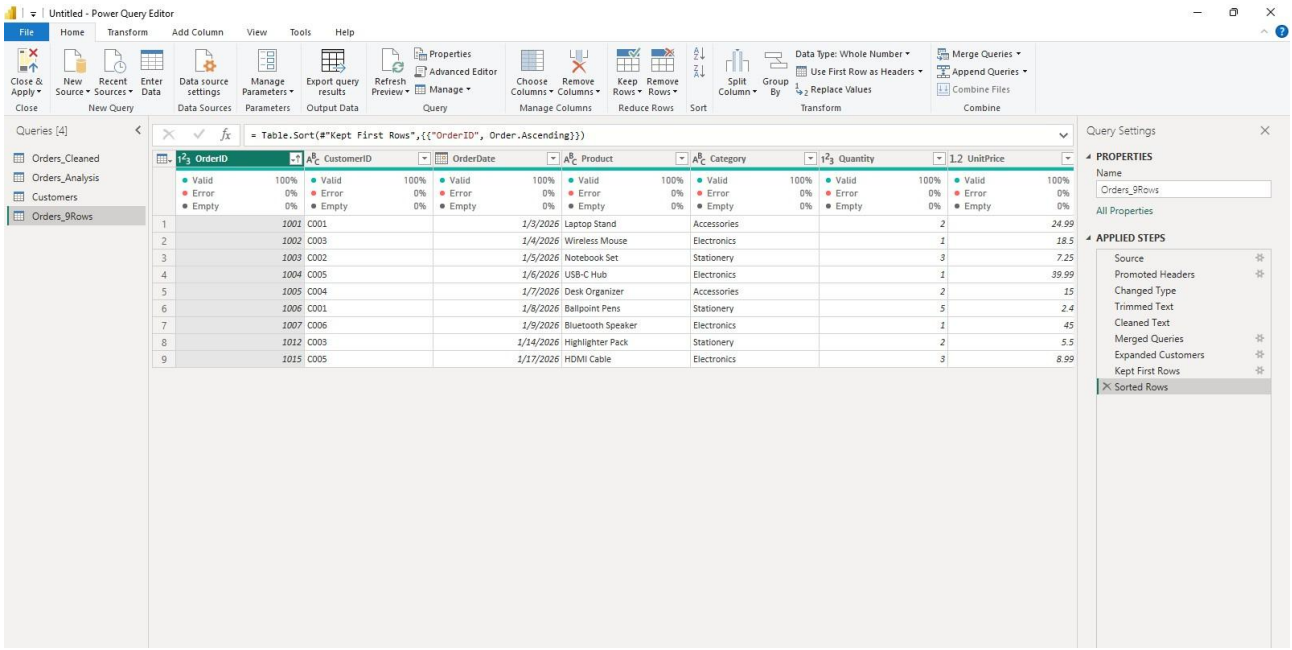
Task 5 : Expand customer columns



	UnitPrice	OrderStatus	CustomerName	City	Email	JoinDate	Segment
1	24.99	Delivered	Aarav Sharma	Delhi	aarav.sharma@example.com	2/14/2025	Regular
2	2.4	Delivered	Aarav Sharma	Delhi	aarav.sharma@example.com	2/14/2025	Regular
3	18.5	Delivered	Kabir Singh	Faridabad	kabir.singh@example.com	11/19/2024	Premium
4	5.5	Delivered	Kabir Singh	Faridabad	kabir.singh@example.com	11/19/2024	Premium
5	7.25	Pending	Diya Verma	Jaipur	diya.verma@example.com	3/2/2025	New
6	39.99	Delivered	Ishaan Patel	Ahmedabad	ishaan.patel@example.com	8/25/2024	Premium
7	8.99	Delivered	Ishaan Patel	Ahmedabad	ishaan.patel@example.com	8/25/2024	Premium
8	15	Cancelled	Anaya Gupta	Lucknow	anaya.gupta@example.com	1/8/2025	Regular
9	45	Pending	Meera Joshi	Pune	meera.joshi@example.com	4/12/2025	New
10	3.25	Delivered	Vivaan Rao	Hyderabad	vivaan.rao@example.com	12/3/2024	Regular
11	12.75	Delivered	Sara Khan	Mumbai	sara.khan@example.com	5/21/2025	Regular
12	54.99	Delivered	Arjun Mehta	Chandigarh	arjun.mehta@example.com	10/17/2024	Premium
13	9.99	Pending	Ira Nair	Bengaluru	ira.nair@example.com	6/9/2025	New
14	32	Delivered	Reyansh Kapoor	Kolkata	reyansh.kapoor@example.com	9/30/2024	Regular
15	19.95	Cancelled	Aadhya Das	Indore	aadhya.das@example.com	7/1/2025	New
16	1.75	Pending	Krishna Malhotra	Surat	krishna.malhotra@example.com	7/16/2024	Premium
17	27.5	Delivered	Myra Chawla	Noida	myra.chawla@example.com	8/11/2025	Regular
18	29.99	Delivered	Sara Khan	Mumbai	sara.khan@example.com	5/21/2025	Regular
19	11.25	Pending	Advik Roy	Bhopal	advik.roy@example.com	9/4/2025	New
20	22	Delivered	Ira Nair	Bengaluru	ira.nair@example.com	6/9/2025	New

Summary: The Customers column was expanded to show CustomerName, City, Email, JoinDate and Segment next to each order. The table still has 20 rows, so no order was lost or repeated. The Expanded Customers step is added in Applied Steps.

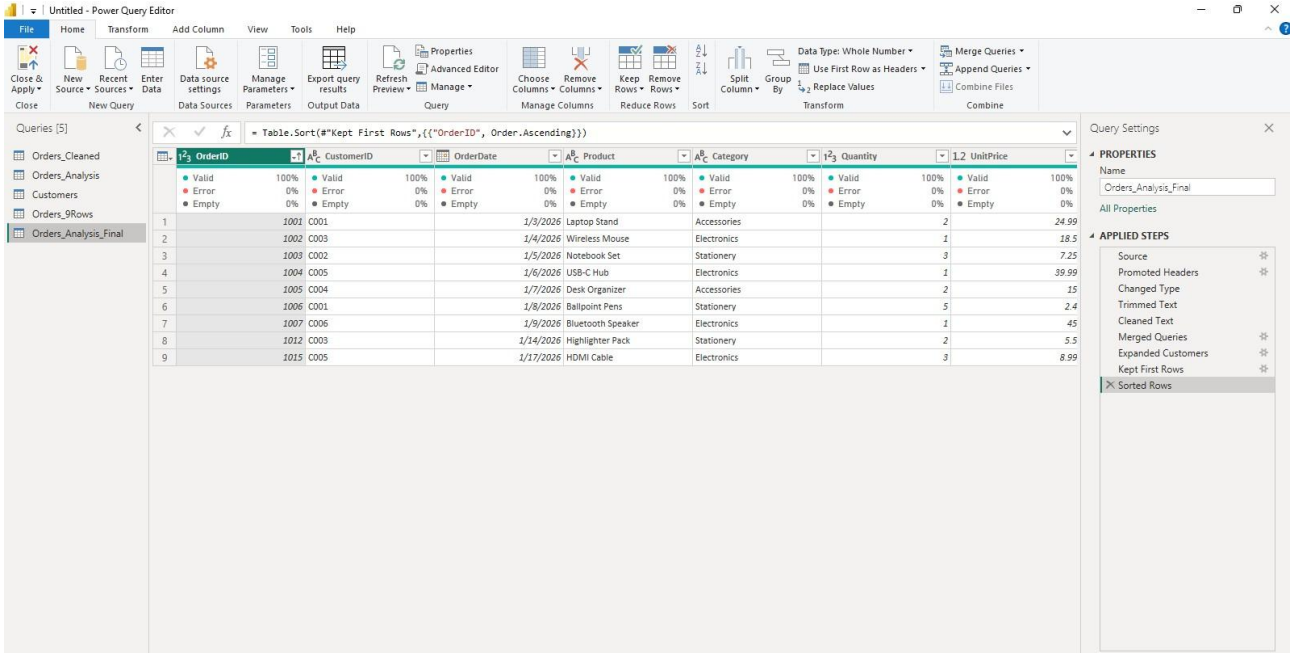
Task 6 : Create the cleaned 9-row table



	OrderID	CustomerID	OrderDate	Product	Category	Quantity	UnitPrice
1	1001	C001	1/3/2026	Laptop Stand	Accessories	2	24.99
2	1002	C003	1/4/2026	Wireless Mouse	Electronics	1	18.5
3	1003	C002	1/5/2026	Notebook Set	Stationery	3	7.25
4	1004	C005	1/6/2026	USB-C Hub	Electronics	1	39.99
5	1005	C004	1/7/2026	Desk Organizer	Accessories	2	15
6	1006	C001	1/8/2026	Ballpoint Pens	Stationery	5	2.4
7	1007	C006	1/9/2026	Bluetooth Speaker	Electronics	1	45
8	1012	C003	1/14/2026	Highlighter Pack	Stationery	2	5.5
9	1015	C005	1/17/2026	HDMI Cable	Electronics	3	8.99

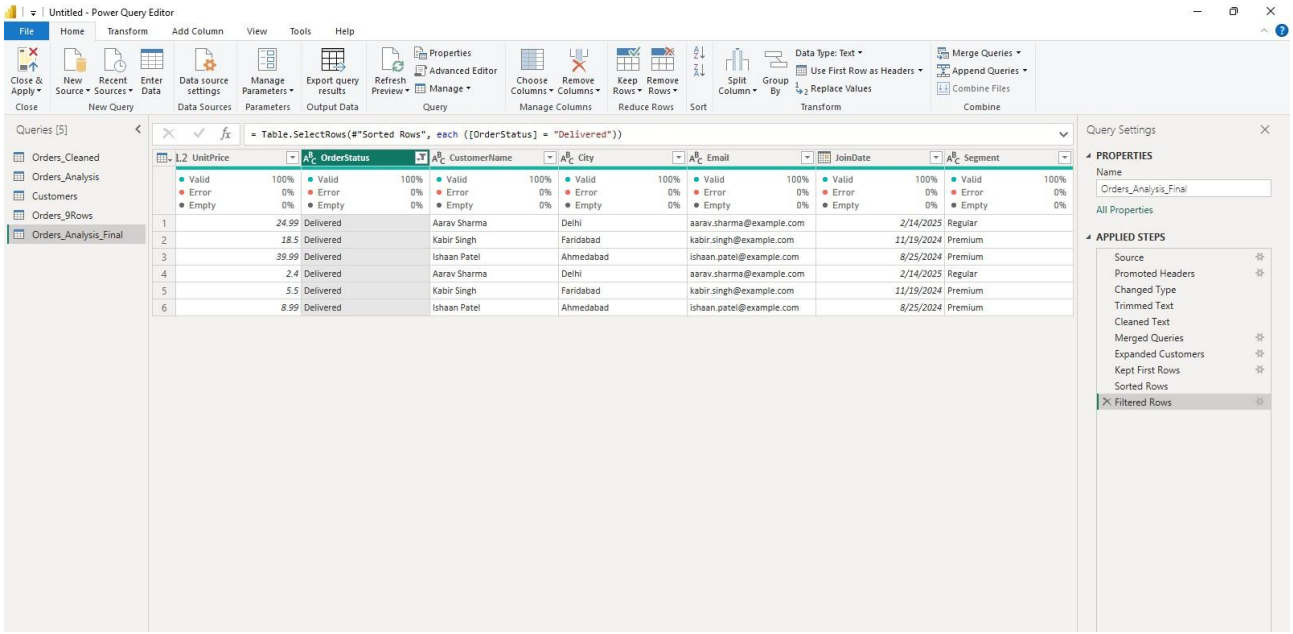
Summary: The merged data was reduced to 9 rows using Keep First Rows and then sorted by OrderID in ascending order. This new query is named Orders_9Rows. It is the cleaned 9-row table, and it is kept unchanged for the rest of the assignment.

Task 7 : Make a copy of the 9-row table



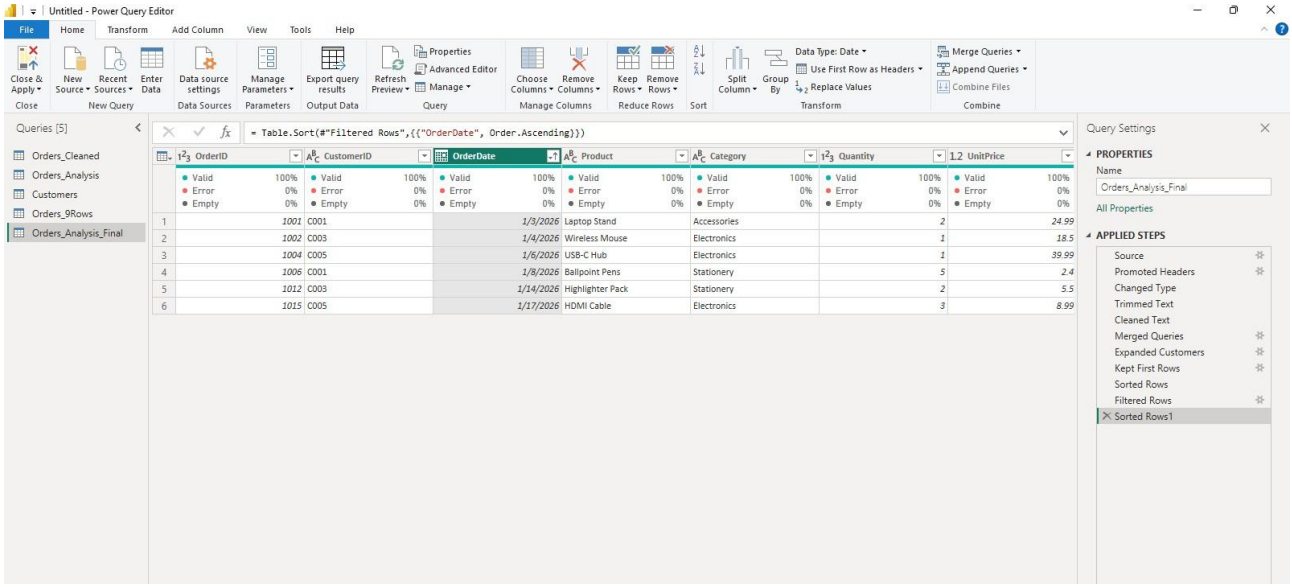
Summary: A copy of the 9-row table was created and named Orders_Analysis_Final. It has the same 9 rows and the same steps as the original. The filter, sort and group tasks are done on this copy so the original table stays unchanged.

Task 8 : Filter rows



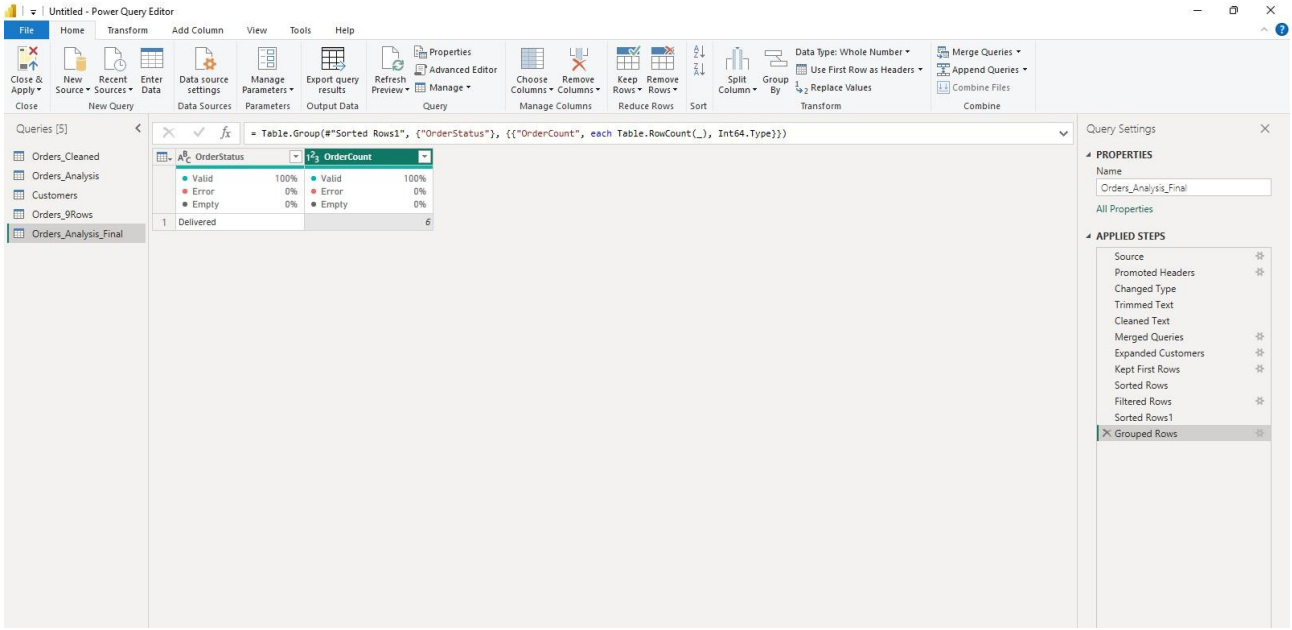
Summary: On the copy, the OrderStatus column was filtered to keep only Delivered orders. The formula bar shows Table.SelectRows with OrderStatus equal to Delivered. The table now has 6 rows, and Pending and Cancelled orders are removed from the copy.

Task 9 : Sort rows



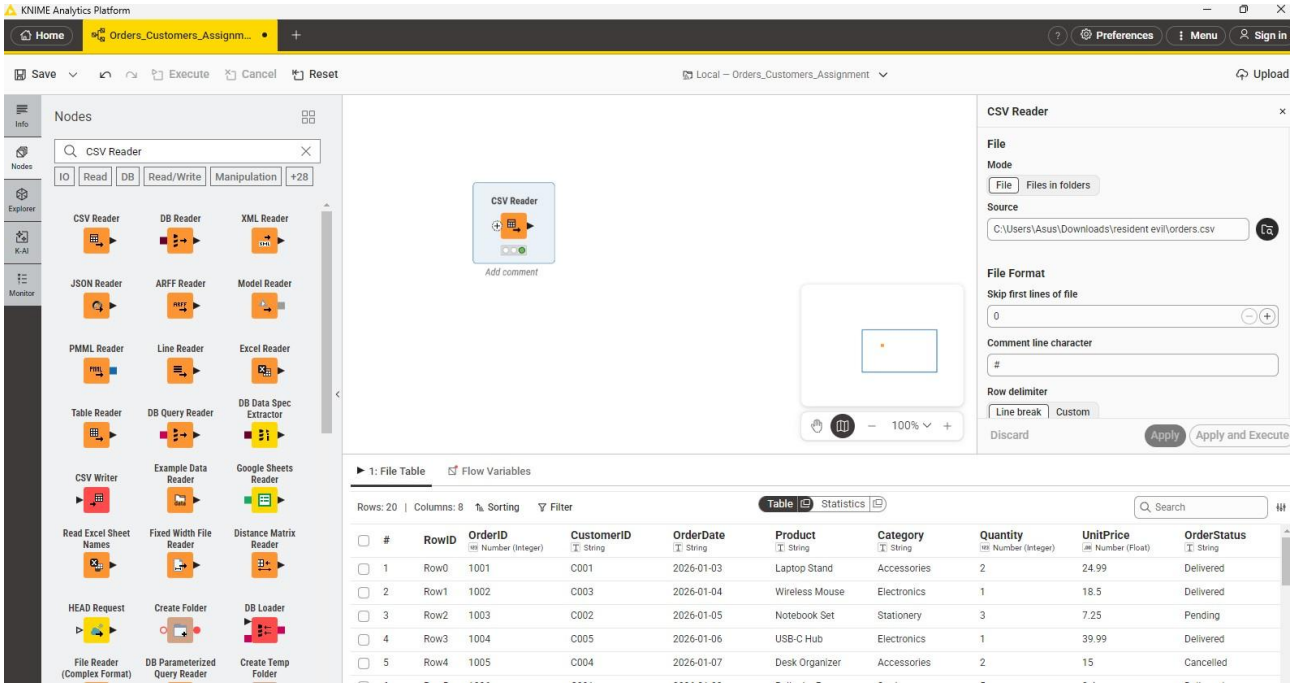
Summary: The filtered data was sorted by OrderDate in ascending order using the Sorted Rows1 step. The 6 delivered orders are now listed from the oldest date to the newest date. The number of rows stays the same because sorting only changes the order.

Task 10 : Group rows



Summary: The Group By option was used on the OrderStatus column to count the rows. A new column called OrderCount was created. The result is one row, Delivered with OrderCount 6, which matches the 6 delivered orders from the filter step.

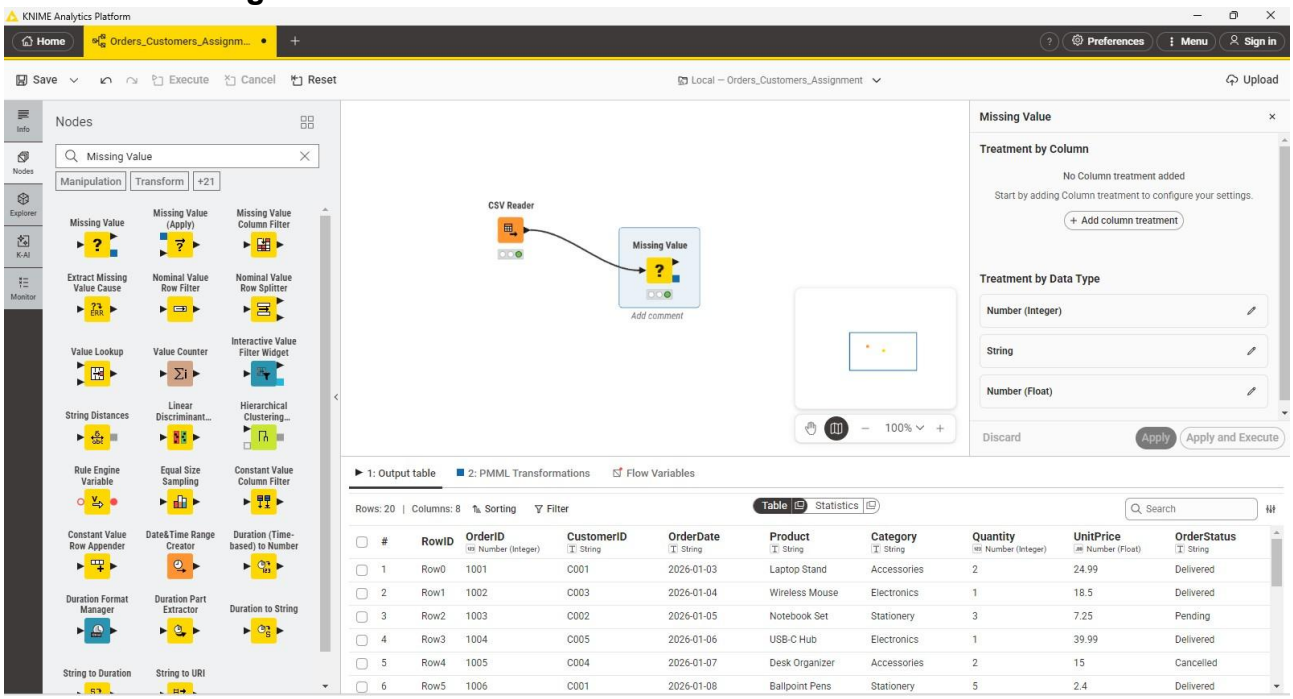
Task 11 : Read orders.csv



#	RowID	OrderID	CustomerID	OrderDate	Product	Category	Quantity	UnitPrice	OrderStatus
1	Row0	1001	C001	2026-01-03	Laptop Stand	Accessories	2	24.99	Delivered
2	Row1	1002	C003	2026-01-04	Wireless Mouse	Electronics	1	18.5	Delivered
3	Row2	1003	C002	2026-01-05	Notebook Set	Stationery	3	7.25	Pending
4	Row3	1004	C005	2026-01-06	USB-C Hub	Electronics	1	39.99	Delivered
5	Row4	1005	C004	2026-01-07	Desk Organizer	Accessories	2	15	Cancelled

Summary: In KNIME, the CSV Reader node was added to the workflow and pointed to orders.csv. The output table shows 20 rows and 8 columns: OrderID, CustomerID, OrderDate, Product, Category, Quantity, UnitPrice and OrderStatus. The green light shows the node ran successfully.

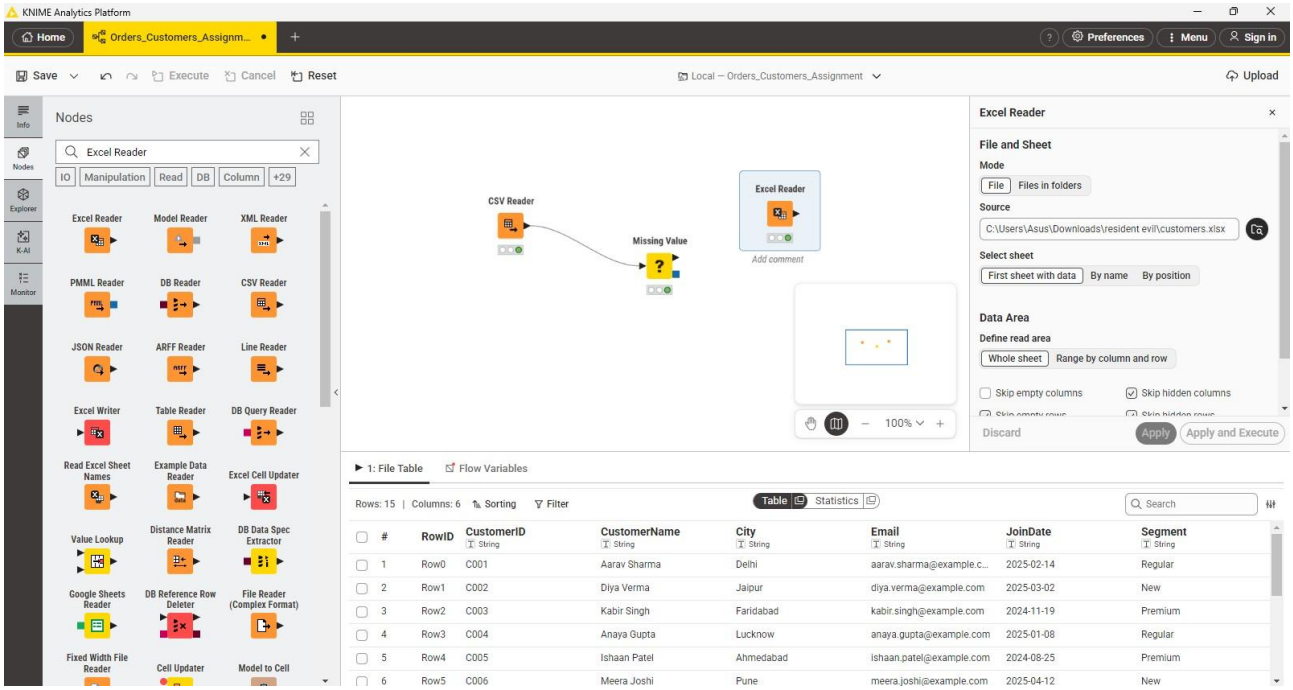
Task 12 : Missing Value node



#	RowID	OrderID	CustomerID	OrderDate	Product	Category	Quantity	UnitPrice	OrderStatus
1	Row0	1001	C001	2026-01-03	Laptop Stand	Accessories	2	24.99	Delivered
2	Row1	1002	C003	2026-01-04	Wireless Mouse	Electronics	1	18.5	Delivered
3	Row2	1003	C002	2026-01-05	Notebook Set	Stationery	3	7.25	Pending
4	Row3	1004	C005	2026-01-06	USB-C Hub	Electronics	1	39.99	Delivered
5	Row4	1005	C004	2026-01-07	Desk Organizer	Accessories	2	15	Cancelled
6	Row5	1006	C001	2026-01-08	Ballpoint Pens	Stationery	5	2.4	Delivered

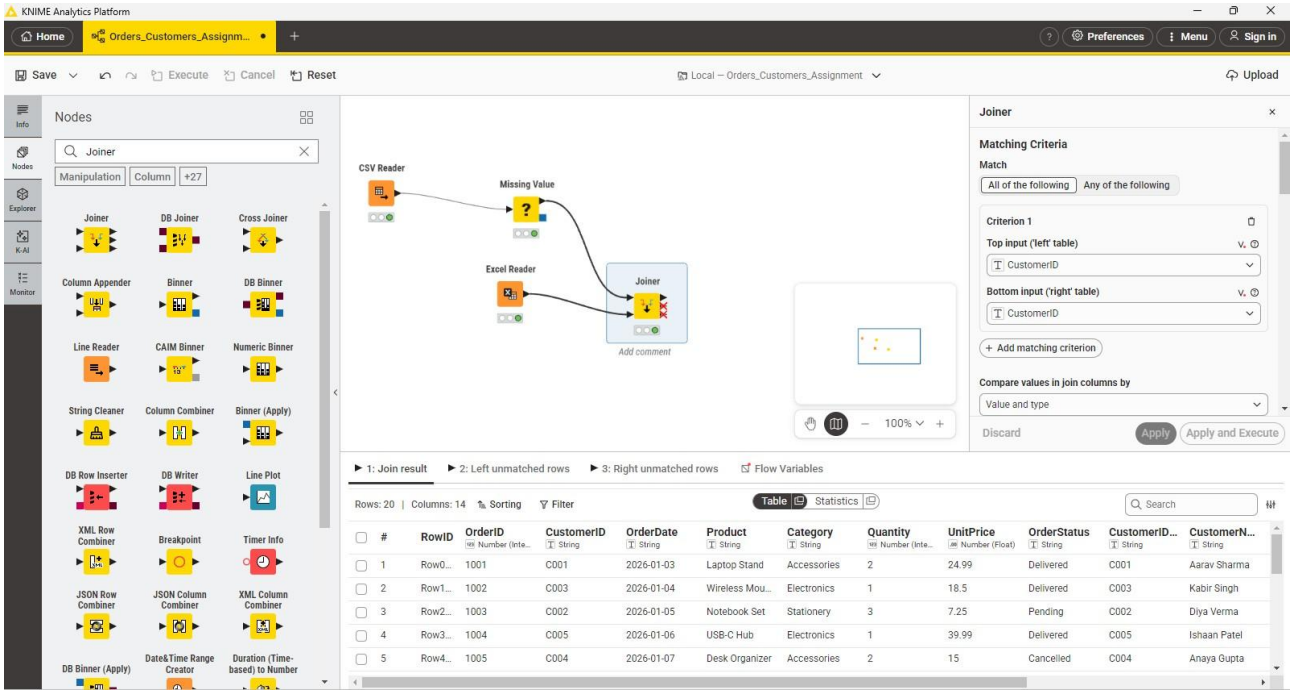
Summary: A Missing Value node was connected to the CSV Reader to check for missing data. The output still shows 20 rows and 8 columns, which means no rows were lost and the orders data has no missing values to fix. The cleaned data now moves to the next step.

Task 13 : Read customers.xlsx



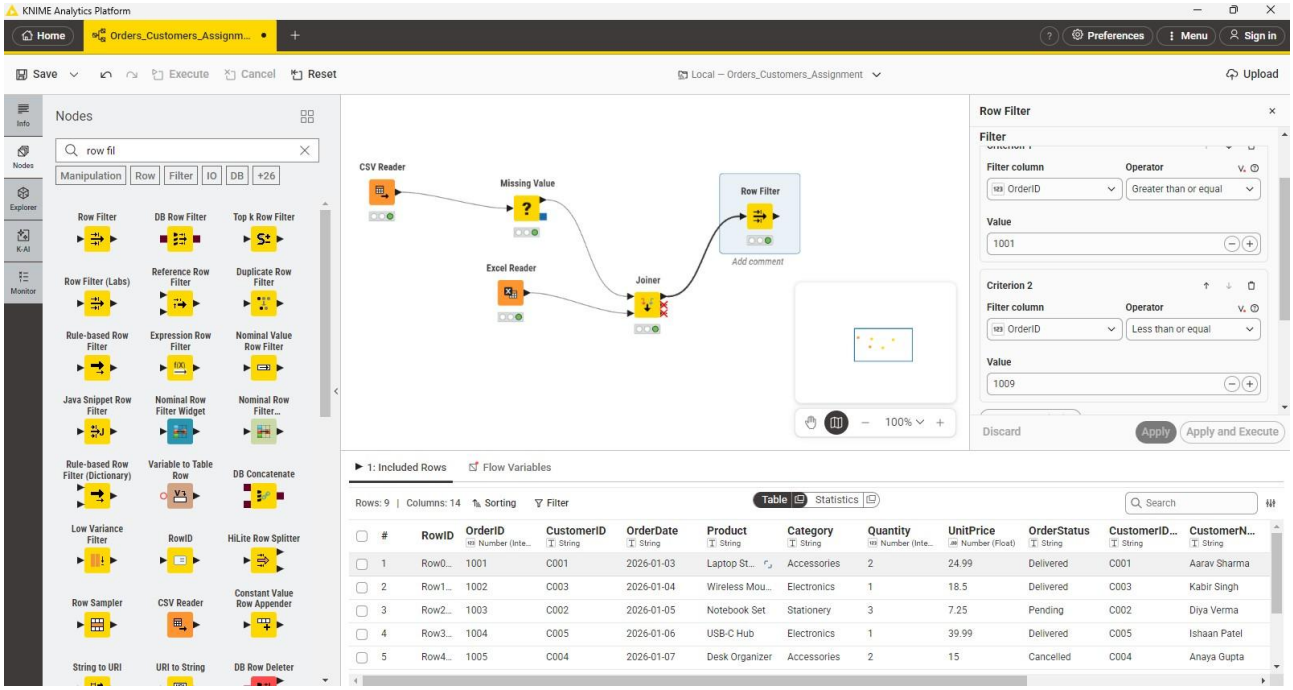
Summary: An Excel Reader node was added on a separate branch and pointed to customers.xlsx. The output has 15 rows and 6 columns: CustomerID, CustomerName, City, Email, JoinDate and Segment. This is the customer data that will be joined with the orders.

Task 14 : Join orders and customers



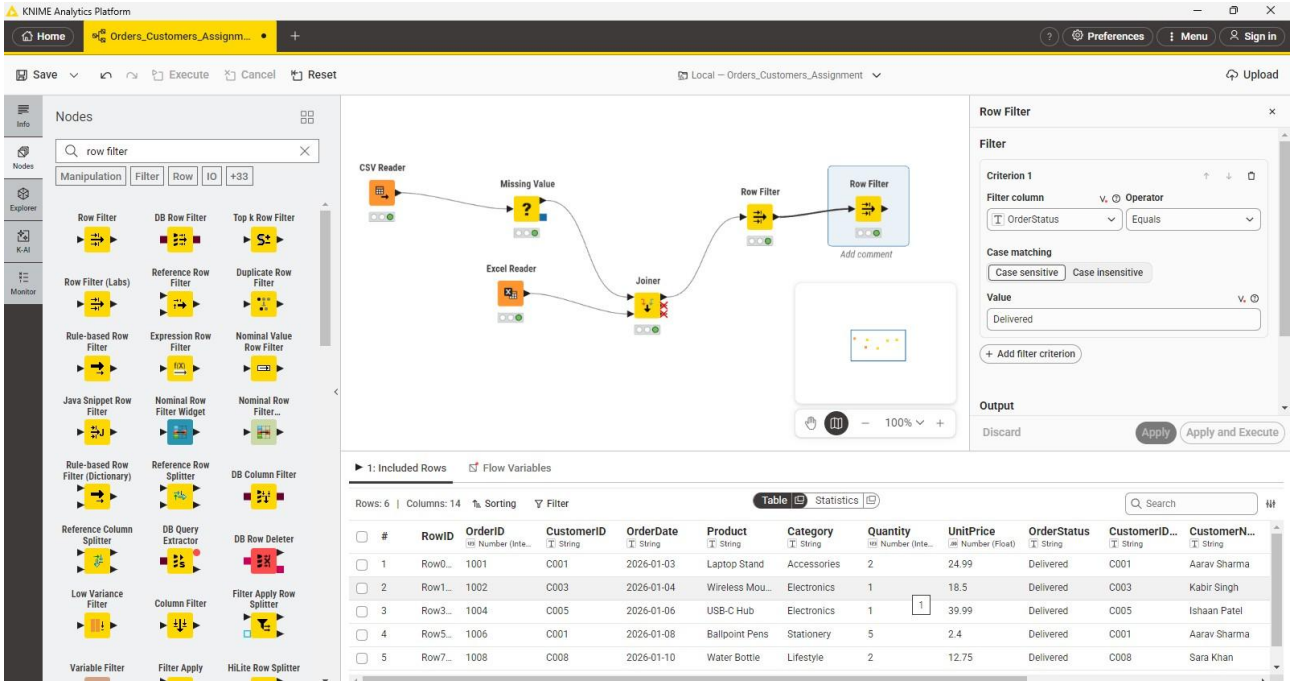
Summary: A Joiner node was used to merge the orders and the customers. The matching column is CustomerID in both the top table and the bottom table. The join result has 20 rows and 14 columns, so every order now has its customer details.

Task 15 : Create the 9-row table



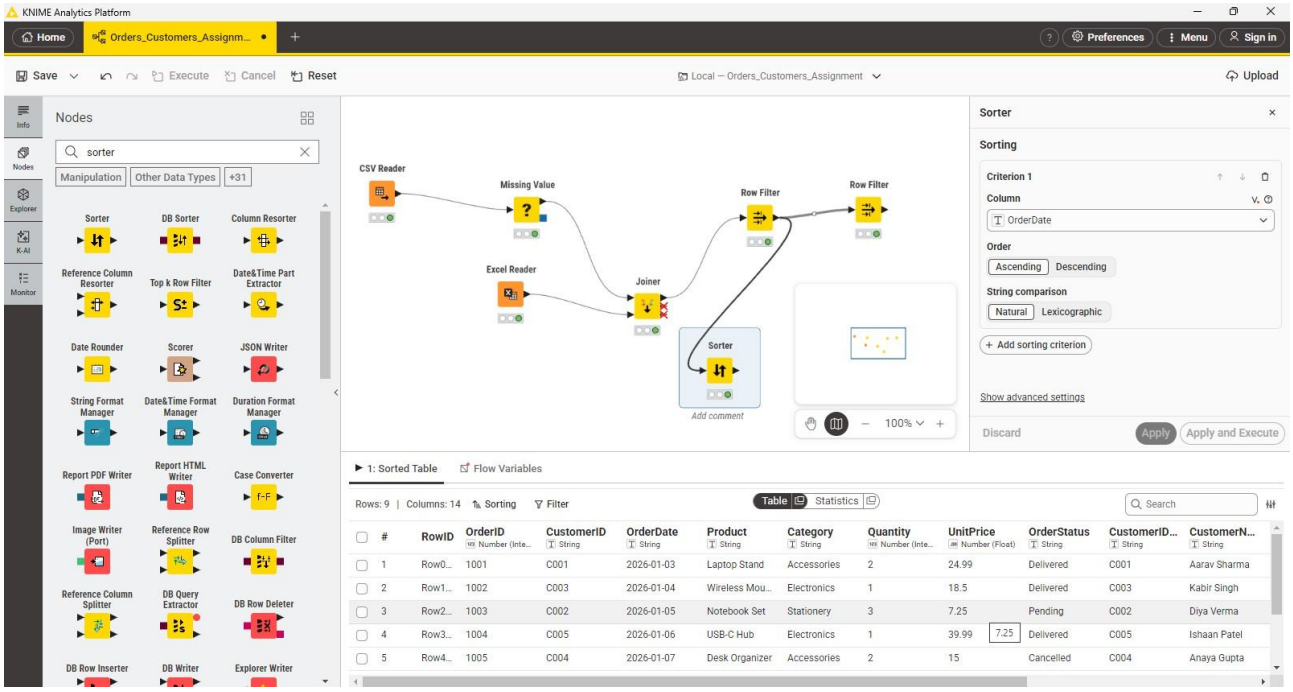
Summary: A Row Filter node was added after the Joiner with two rules on OrderID: greater than or equal to 1001 and less than or equal to 1009. The output has 9 rows and 14 columns. This is the 9-row table, and it is not changed again because the next nodes branch from it.

Task 16 : Filter rows



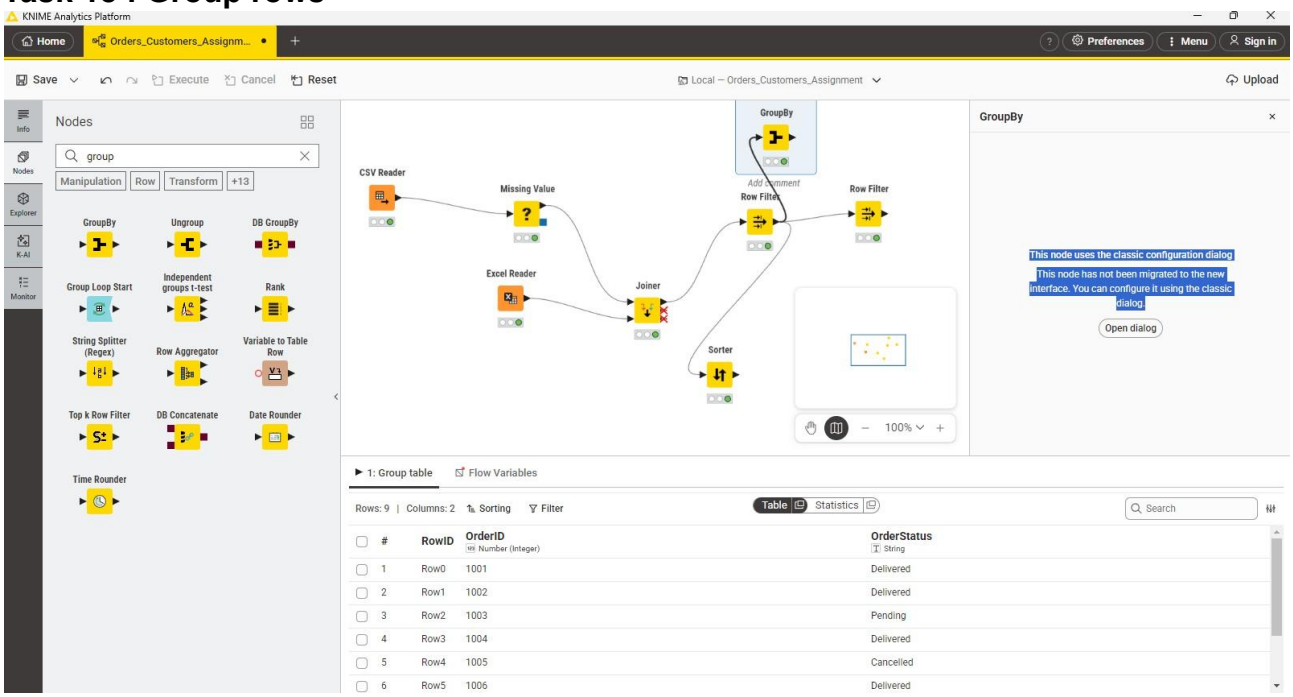
Summary: A second Row Filter node was connected to the 9-row table. Its rule is OrderStatus equals Delivered. The output shows 6 rows and all of them have the status Delivered. This is done on a branch, so the 9-row table is unchanged.

Task 17 : Sort rows



Summary: A Sorter node was added on a separate branch from the 9-row table. It sorts the data by OrderDate in ascending order. The output has 9 rows listed from the earliest date to the latest date, and the original 9-row table is not changed.

Task 18 : Group rows



Summary: A GroupBy node was added and executed. The group table has 2 columns, OrderID and OrderStatus, and 9 rows. All nodes in the workflow show a green light, which means the whole workflow ran successfully.